Ice Dynamics I: Strain Rates, Simple Shear, and Ice Sheet Profiles

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Topics Covered

- Strain-rate tensor and the velocity gradient
- Glen's flow law (stated; derivation and physical basis in a later lecture)
- Simple shear: the parallel slab on a slope
- The shallow-ice approximation (SIA)
- Ice sheet surface profiles: the Vialov solution and the perfectly plastic limit
- Comparison with observed ice sheet profiles

Primary reference: Cuffey & Paterson, *Physics of Glaciers*, 4th ed., Chapters 3 and 8–9.

1 Motivation and Review

In the preceding lectures we established:

1. **Mass balance** provides the climate forcing. The ice-equivalent specific balance rate $\dot{b}_i$ enters the vertically integrated continuity equation,

$$\frac{\partial h}{\partial t} = \dot{b}_i - \frac{\partial q}{\partial x}, \tag{1}$$

   where $q = \bar{u}\, h$ is the depth-integrated volume flux per unit width.

2. **Stress analysis** gives the internal mechanical state. In particular, the *driving stress*,

$$\tau_d = -\rho_i g\, h\, \frac{\partial s}{\partial x}, \tag{2}$$

   quantifies the gravitational force per unit area that propels the ice forward, and the *deviatoric stress tensor* τ_{ij}, defined through $\sigma_{ij} = -p\,\delta_{ij} + \tau_{ij}$, isolates the stress components that cause ice to deform.

What is still missing is a way to compute the velocity field. For that we need two additional ingredients:

- A **kinematic** measure of deformation—the strain-rate tensor.
- A **constitutive relation** linking stress to deformation rate—Glen's flow law.

Plan for this lecture. We introduce the strain-rate tensor (§2), state Glen's flow law (§3), solve the simplest glacier-flow problem—simple shear of a parallel slab (§4)—and then use that solution to build ice sheet surface profiles from first principles (§5–7).

2 Strain Rates and the Velocity Gradient Tensor

2.1 The Velocity Gradient Tensor

Let $u_i(\boldsymbol{x}, t)$ denote the velocity field in the ice. The **velocity gradient tensor** is

$$L_{ij} = \frac{\partial u_i}{\partial x_j}. \tag{3}$$

This tensor contains all information about how the velocity field varies in space at a given point. It can be decomposed uniquely into symmetric and antisymmetric parts:

$$L_{ij} = \dot{\varepsilon}_{ij} + \omega_{ij}, \tag{4}$$

where

$$\dot{\varepsilon}_{ij} = \tfrac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \qquad \text{(strain-rate tensor)}, \tag{5}$$

$$\omega_{ij} = \tfrac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \qquad \text{(spin tensor)}. \tag{6}$$

The **strain-rate tensor** $\dot{\varepsilon}_{ij}$ measures the rate at which material elements stretch, compress, and shear. The **spin tensor** ω_{ij} measures the local rate of rigid-body rotation; it does not contribute to deformation and does not enter the constitutive law.

2.2 Physical Interpretation of the Components

The strain-rate tensor is symmetric ($\dot{\varepsilon}_{ij} = \dot{\varepsilon}_{ji}$) and therefore has six independent components in three dimensions.

Diagonal components ($\dot{\varepsilon}_{xx}$, $\dot{\varepsilon}_{yy}$, $\dot{\varepsilon}_{zz}$), aka the normal strain rates, are the rates of stretching or compression along each coordinate axis. For example:

$$\dot{\varepsilon}_{xx} = \frac{\partial u}{\partial x} \tag{7}$$

is the rate at which a material element elongates (if positive) or shortens (if negative) in the x-direction.

Off-diagonal components ($\dot{\varepsilon}_{xz}$, $\dot{\varepsilon}_{xy}$, $\dot{\varepsilon}_{yz}$) measure shear deformation—the rate at which originally perpendicular material lines rotate toward or away from each other. For example:

$$\dot{\varepsilon}_{xz} = \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) \tag{8}$$

measures shearing in the x–z plane.

In a glacier context:

- $\dot{\varepsilon}_{xz}$: vertical shearing of the horizontal velocity—the dominant mode of deformation in slowly flowing ice sheets (the focus of this lecture).
- $\dot{\varepsilon}_{xx}$: longitudinal stretching or compression—important near termini, ice divides, and in ice streams where velocity changes rapidly along flow.
- $\dot{\varepsilon}_{xy}$: lateral shearing—important at the margins of ice streams and outlet glaciers.

2.3 Incompressibility

Below the firn layer, glacier ice has approximately constant density ($\rho_i \approx 917\,\mathrm{kg\,m^{-3}}$). Conservation of mass then requires that the velocity field be divergence-free:

$$\boxed{\dot{\varepsilon}_{kk} = \frac{\partial u_i}{\partial x_i} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.} \tag{9}$$

This is the **incompressibility constraint**: any stretching in one direction must be compensated by compression in another. Only two of the three normal strain rates are independent.

Example. If ice is stretching longitudinally ($\dot{\varepsilon}_{xx} > 0$) with no lateral strain ($\dot{\varepsilon}_{yy} = 0$), then the ice must be thinning vertically: $\dot{\varepsilon}_{zz} = -\dot{\varepsilon}_{xx} < 0$.

Derivation: Incompressibility of Glacier Ice

We derive Eq. (9) from conservation of mass. The general continuity equation for a material with density ρ and velocity field u_i is

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho\, u_i) = 0.$$

Expanding the divergence term using the product rule:

$$\frac{\partial \rho}{\partial t} + u_i \frac{\partial \rho}{\partial x_i} + \rho\, \frac{\partial u_i}{\partial x_i} = 0.$$

The first two terms together form the material (Lagrangian) derivative of density:

$$\frac{D\rho}{Dt} + \rho\, \frac{\partial u_i}{\partial x_i} = 0.$$

Below the firn layer, the density of glacier ice is effectively constant ($\rho \approx \rho_i = 917\ \mathrm{kg\,m^{-3}}$), meaning the density of each material parcel does not change as it moves: $D\rho/Dt = 0$. Therefore:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \boldsymbol{u} = 0.$$

Since the trace of the strain-rate tensor is $\dot{\varepsilon}_{kk} = \partial u_i/\partial x_i$ (the antisymmetric part ω_{ij} is traceless and does not contribute), we recover Eq. (9):

$$\boxed{\dot{\varepsilon}_{kk} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.}$$

2.4 Effective Strain Rate

By analogy with the effective deviatoric stress τ_e from the stress lecture, we define the **effective strain rate**:

$$\dot{\varepsilon}_e = \sqrt{\frac{\dot{\varepsilon}_{ij}\,\dot{\varepsilon}_{ij}}{2}}. \tag{10}$$

This is a scalar invariant that measures the overall intensity of deformation regardless of orientation. It will appear in Glen's flow law. Expanding the repeated-index contraction $\dot{\varepsilon}_{ij}\,\dot{\varepsilon}_{ij}$:

$$\dot{\varepsilon}_e = \sqrt{\dot{\varepsilon}_{xx}^2 + \dot{\varepsilon}_{yy}^2 + \dot{\varepsilon}_{zz}^2 + 2\dot{\varepsilon}_{xy}^2 + 2\dot{\varepsilon}_{xz}^2 + 2\dot{\varepsilon}_{yz}^2}. \tag{11}$$

The factors of 2 on the shear terms arise because $\dot{\varepsilon}_{ij}\,\dot{\varepsilon}_{ij}$ sums over all nine components and the off-diagonal terms appear twice by symmetry ($\dot{\varepsilon}_{xy} = \dot{\varepsilon}_{yx}$, etc.).

3 Glen's Flow Law (Stated, later derived and discussed)

The constitutive relation for the creep of polycrystalline ice relates the deviatoric stress to the strain rate:

$$\boxed{\dot{\varepsilon}_{ij} = A\,\tau_e^{n-1}\,\tau_{ij},} \tag{12}$$

where:

- A is the **rate factor** (units $\mathrm{Pa}^{-n}\,\mathrm{s}^{-1}$), which depends strongly on temperature through an Arrhenius relation. For now we treat A as a known constant; its physical basis is the subject of a later lecture.
- $n \approx 3$ is the **creep exponent**.
- $\tau_e = \sqrt{\tau_{ij}\,\tau_{ij}/2}$ is the effective deviatoric stress.

The key feature of Glen's law is its **nonlinearity**: because $n > 1$, the effective viscosity

$$\eta = \frac{1}{2A\,\tau_e^{n-1}} \tag{13}$$

decreases with increasing stress. Ice is a *shear-thinning* fluid: at high stress (e.g., near the bed of a glacier), the ice is "softer" and deforms more readily. For $n = 3$, doubling the stress increases the strain rate by a factor of 8.

The concentration of deformation in high-stress regions is a recurring theme: it explains why most of the shear in an ice sheet occurs near the bed, and why ice streams and outlet glaciers can flow orders of magnitude faster than their surroundings.

4 Simple Shear: The Parallel Slab

The simplest problem in glacier dynamics is an infinite, uniform slab of ice resting on an inclined plane. Despite its simplicity, this problem captures the dominant physics of ice-sheet interiors and provides the velocity profile used throughout glaciology.

4.1 Setup and Assumptions

Consider a slab of ice with uniform thickness h on a bed inclined at angle α to the horizontal. We adopt coordinates with x pointing downslope and z pointing upward from the bed, so the bed is at $z = b$ and the surface at $z = s = b + h$.

Assumptions:

1. The geometry is uniform along flow: $\partial/\partial x = 0$ for all quantities except the velocity itself.

2. The flow is two-dimensional: no variation in y.
3. The ice is in mechanical equilibrium (no acceleration).
4. The upper surface is stress-free (no wind or atmospheric loading).

Under these assumptions, the only nonzero strain rate is $\dot{\varepsilon}_{xz}$, and the only relevant stress component driving deformation is the shear stress τ_{xz}.

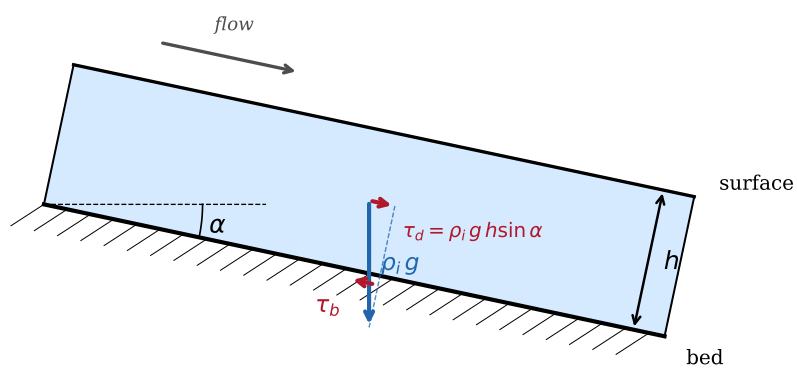

Figure 1: Parallel-slab geometry. A uniform slab of ice with thickness h rests on a bed inclined at angle α. The free body extends from the surface down to depth $d = s - z$: the downslope gravity component $\rho_i g d \sin\alpha$ is balanced by the shear traction $|\tau_{xz}(z)|$ on the base of the block.

4.2 Stress Distribution from a Free-Body Diagram

Consider a block of ice extending from the free surface ($z = s$) down to some level z within the ice. The block has thickness $d = s - z$ and unit horizontal and lateral extent.

The forces acting on this block per unit area are:

- **Gravity component along the slope:** $\rho_i g\, d\, \sin\alpha$ (downslope).
- **Shear traction on the base of the block:** $|\tau_{xz}(z)|$ (opposing motion, upslope).
- **Tractions on the surface:** zero (free surface).

Balancing forces in the downslope ($+x$) direction. The gravity component is $\rho_i g\, d\, \sin\alpha$ (downslope). The shear traction on the base of the block, exerted by the ice below on the face with outward normal $-\hat{z}$, is $-\tau_{xz}$ (the minus sign from the outward normal). Setting the net force to zero:

$$\rho_i g\,(s - z)\,\sin\alpha - \tau_{xz}(z) = 0, \tag{14}$$

so:

$$\boxed{\tau_{xz}(z) = \rho_i g\,(s - z)\,\sin\alpha.} \tag{15}$$

The shear stress is positive: ice above height z is driven downslope by gravity, and the resulting viscous coupling means faster-moving ice above drags slower-moving ice below. The stress increases linearly from zero at the surface to a maximum at the bed:

$$\tau_b \equiv \tau_{xz}(b) = \rho_i g\, h\, \sin\alpha. \tag{16}$$

For small slopes ($\sin\alpha \approx \tan\alpha = -\partial s/\partial x$), the basal shear stress equals the driving stress:

$$\tau_b = \rho_i g\, h\, \sin\alpha \approx -\rho_i g\, h\, \frac{\partial s}{\partial x} = \tau_d, \tag{17}$$

recovering the expression from the stress lecture.

Note: We have derived this stress distribution from a simple force balance on a block of ice. This is a special case of the general conservation of linear momentum, which we will introduce formally in a later lecture. The key simplification here is that $\partial/\partial x = 0$ eliminates all terms involving longitudinal and lateral stress gradients.

4.3 Velocity Profile

Derivation: Velocity Profile for Simple Shear

With only τ_{xz} nonzero, the effective stress simplifies to $\tau_e = \tau_{xz}$ (which is positive). Glen's flow law (12) gives:

$$\dot{\varepsilon}_{xz} = A\, \tau_{xz}^{n-1}\, \tau_{xz} = A\, \tau_{xz}^{n}. \tag{18}$$

Since $\partial w/\partial x = 0$ for the parallel slab, the shear strain rate is $\dot{\varepsilon}_{xz} = \frac{1}{2}\partial u/\partial z$, so:

$$\frac{\partial u}{\partial z} = 2A\left(\rho_i g \sin\alpha\right)^n (s-z)^n. \tag{19}$$

Integrating from the bed ($z = b$, where $u = u_b$) to height z. Let $\zeta = s - z'$ so that $d\zeta = -dz'$:

$$\begin{aligned} u(z) - u_b &= \int_b^z 2A\,(\rho_i g \sin\alpha)^n (s-z')^n\, dz' \\ &= 2A\,(\rho_i g \sin\alpha)^n \left[-\frac{(s-z')^{n+1}}{n+1}\right]_b^z \\ &= \frac{2A}{n+1}\,(\rho_i g \sin\alpha)^n \left[h^{n+1} - (s-z)^{n+1}\right]. \end{aligned} \tag{20}$$

The velocity at any height z above the bed is therefore

$$\boxed{u(z) = u_b + \frac{2A}{n+1}\,(\rho_i g \sin\alpha)^n \left[h^{n+1} - (s-z)^{n+1}\right],} \tag{21}$$

where u_b is the basal sliding velocity (prescribed as a boundary condition; for now we set $u_b = 0$).

The surface velocity (at $z = s$) is:

$$u_s = u_b + \frac{2A}{n+1}\,(\rho_i g \sin\alpha)^n\, h^{n+1}. \tag{22}$$

The velocity due to internal deformation scales as $\tau_d^n\, h$, or equivalently as h^{n+1} for fixed surface slope. For $n = 3$, the deformational velocity depends on the *fourth power* of the ice thickness—a very strong sensitivity.

4.4 Shape of the Velocity Profile

Figure 2 shows normalized velocity profiles for several values of n. The key observation is that deformation is **concentrated near the bed**. Writing $u(z)/u_s$ as a function of the normalized height $\hat{z} = (z - b)/h$:

$$\frac{u(z)}{u_s} = 1 - (1 - \hat{z})^{n+1}. \tag{23}$$

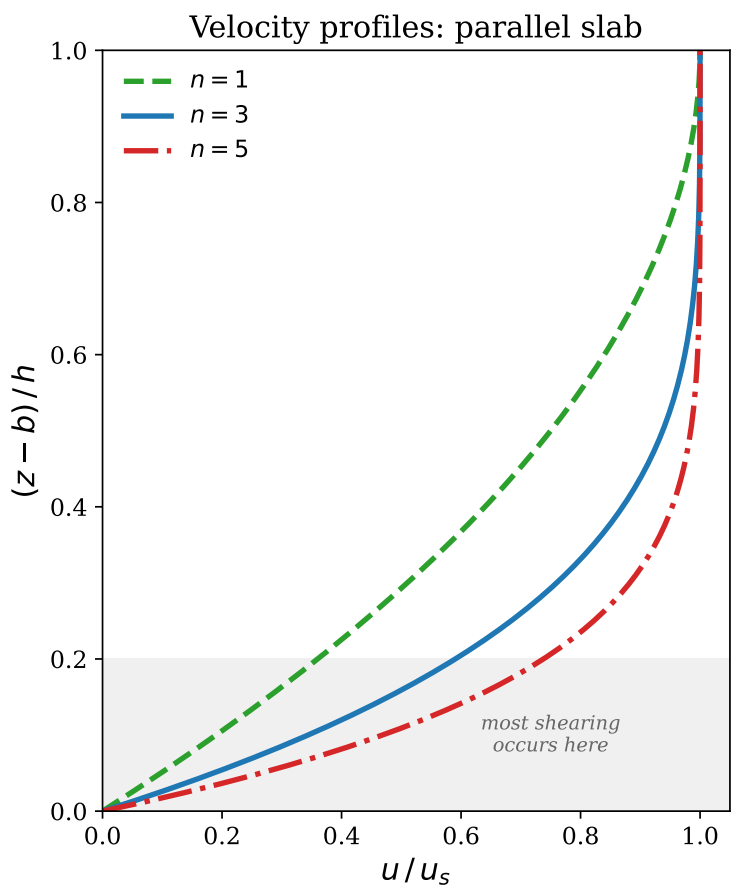

Figure 2: Normalized velocity profiles u/u_s vs. height above bed $(z - b)/h$ for the parallel-slab problem with $n = 1$ (Newtonian), $n = 3$ (standard Glen's law), and $n = 5$. For $n = 3$, roughly 80% of the surface velocity is achieved in the upper half of the ice column; most shearing occurs in the basal 20%.

For $n = 1$ (Newtonian fluid), the profile is parabolic. For $n = 3$, the upper ∼80% of the ice column moves nearly as a rigid "plug," with intense shearing confined to the basal ∼20%. As n increases, the profile becomes increasingly plug-like, approaching a perfectly plastic limit in which all deformation occurs in a thin basal layer.

4.5 Depth-Averaged Velocity and Flux

The depth-averaged velocity is:

$$\bar{u} = \frac{1}{h} \int_b^s u(z)\, dz = u_b + \frac{2A}{n+2} (\rho_i g \sin\alpha)^n h^{n+1}. \tag{24}$$

The ratio of the depth-averaged to the surface deformational velocity is:

$$\frac{\bar{u} - u_b}{u_s - u_b} = \frac{n+1}{n+2}. \tag{25}$$

For $n = 3$: $\bar{u} = u_b + \frac{4}{5}(u_s - u_b)$.

The depth-integrated **flux per unit width** is:

$$\boxed{q = \bar{u}\, h = u_b\, h + \frac{2A}{n+2}\,(\rho_i g \sin\alpha)^n\, h^{n+2}.} \tag{26}$$

This is the expression that will enter the continuity equation to determine ice sheet profiles. Note the strong dependence on thickness: $q \propto h^{n+2}$ for the deformational part (i.e., $q \propto h^5$ for $n = 3$).

Two end-member flow regimes:

- $u_b = 0$: all motion is from internal deformation (frozen bed). This is the regime we focus on for deriving ice sheet profiles.
- $u_b \gg u_s - u_b$: sliding dominates and the flow is approximately plug-like. This regime characterizes ice streams.

5 The Ice Sheet as a Viscous Gravity Current

We now use the slab solution to compute the surface profile of a two-dimensional ice sheet in steady state. The idea is simple: if the ice sheet is much wider than it is thick, then the flow at any point is well approximated by the local slab solution.

5.1 The Shallow-Ice Approximation

A typical large ice sheet has a thickness of $h \sim 3$ km and a half-span of $L \sim 1000$ km, giving an aspect ratio

$$\varepsilon = \frac{[h]}{[L]} \sim 3 \times 10^{-3}. \tag{27}$$

When $\varepsilon \ll 1$, a scaling analysis of the full stress-balance equations (which we will develop in a later lecture) shows that:

- The dominant stress is the vertical shear stress τ_{xz}.
- Longitudinal deviatoric stresses (τ_{xx}) are smaller by a factor of $O(\varepsilon^2)$.
- The pressure is approximately glaciostatic.

Under these conditions, the ice at each location deforms as though it were part of a parallel slab with the local thickness and surface slope. This is the **shallow-ice approximation** (SIA).

5.2 The SIA Flux

Under the SIA with no basal sliding and a flat bed ($s = h$), the flux from Eq. (26) becomes:

$$q(x) = -\frac{2A}{n+2}\,(\rho_i g)^n \left|\frac{\partial s}{\partial x}\right|^{n-1} \frac{\partial s}{\partial x}\, h^{n+2}, \tag{28}$$

where the sign ensures that flux is in the direction of decreasing surface elevation. Substituting $s = h$ (flat bed):

$$q(x) = -\frac{2A}{n+2}\,(\rho_i g)^n \left|\frac{dh}{dx}\right|^{n-1} \frac{dh}{dx}\, h^{n+2}. \tag{29}$$

5.3 Coupling to the Continuity Equation

The vertically integrated continuity equation at steady state is:

$$\dot{b}_i = \frac{dq}{dx}. \tag{30}$$

Substituting the SIA flux (29) yields a nonlinear ordinary differential equation for $h(x)$. In full generality this must be solved numerically, but for uniform accumulation on a flat bed, an analytical solution exists.

5.4 The Vialov Profile

Derivation: The Vialov Profile

Setup. Consider a symmetric ice sheet on a flat, horizontal bed with uniform accumulation rate $\dot{b}_i > 0$ and no basal sliding. Place the origin at the ice divide; the margin is at $x = L$ (half-span). By symmetry, $q(0) = 0$ and $dh/dx\big|_{x=0} = 0$.

Step 1: Integrate the continuity equation.
From Eq. (30), integrating from the divide:

$$q(x) = \dot{b}_i\, x. \tag{31}$$

The flux increases linearly from zero at the divide to $\dot{b}_i L$ at the margin.

Step 2: Equate with the SIA flux.
Using $q > 0$ for $x > 0$ (flow toward the margin, $dh/dx < 0$):

$$\frac{2A}{n+2}\,(\rho_i g)^n \left(-\frac{dh}{dx}\right)^n h^{n+2} = \dot{b}_i\, x. \tag{32}$$

Step 3: Separate variables and integrate.
Rearranging and writing $(-dh/dx)^n = \dot{b}_i x \cdot (n+2)/[2A\,(\rho_i g)^n\, h^{n+2}]$, then integrating with the boundary condition $h(L) = 0$, the solution is:

$$\boxed{h(x) = h_0 \left[1 - \left(\frac{x}{L}\right)^{\frac{n+1}{n}}\right]^{\frac{n}{2(n+1)}},} \tag{33}$$

where the divide thickness is

$$\boxed{h_0 = \left[\frac{2^{n-1}(n+2)\,\dot{b}_i}{A\,(\rho_i g)^n}\right]^{1/(2n+2)} L^{(n+1)/(2n+2)}.} \tag{34}$$

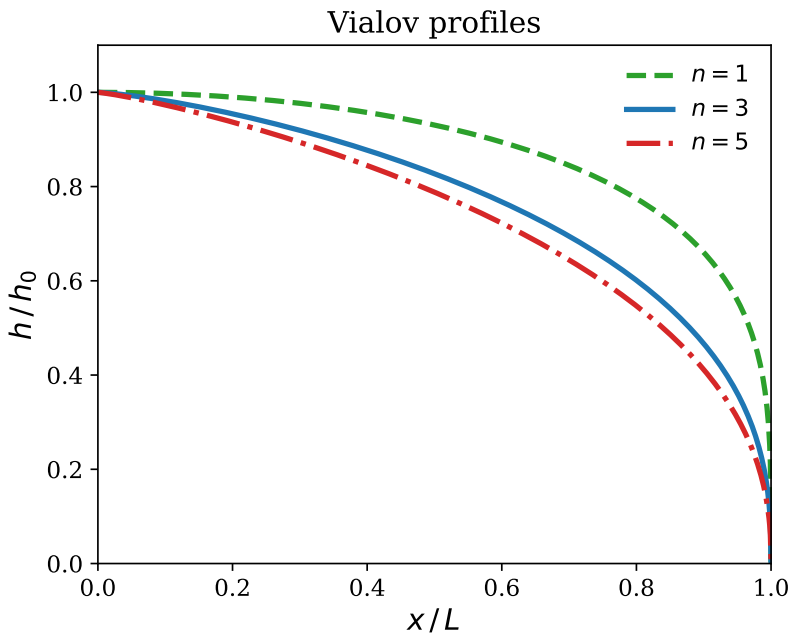

Figure 3: Normalized Vialov profiles h/h_0 vs. x/L for $n = 1$, 3, and 5. Higher n produces steeper margins and a flatter interior, reflecting the increasing concentration of deformation in high-stress regions.

5.5 Insensitivity to Parameters

A remarkable feature of the Vialov solution is the **weak dependence** of h_0 on both the accumulation rate and the rate factor:

$$h_0 \propto \left(\frac{\dot{b}_i}{A}\right)^{1/(2n+2)}. \tag{35}$$

For $n = 3$, the exponent is $1/8 = 0.125$. Consequences:

- **Doubling** $\dot{b}_i$ increases h_0 by only $\sim$9%.
- An **eight-fold** change in $\dot{b}_i$ changes h_0 by $\sim$30%.
- The same insensitivity holds for A, meaning that large uncertainties in the rate factor have a modest effect on the predicted profile.

Ice sheet shape is set primarily by the span L and by gravity; the rheological and climate details enter only weakly. This is a direct consequence of the nonlinearity of Glen's law: the strong dependence of flux on thickness ($q \propto h^{n+2}$) acts as a powerful negative feedback that limits thickness variations.

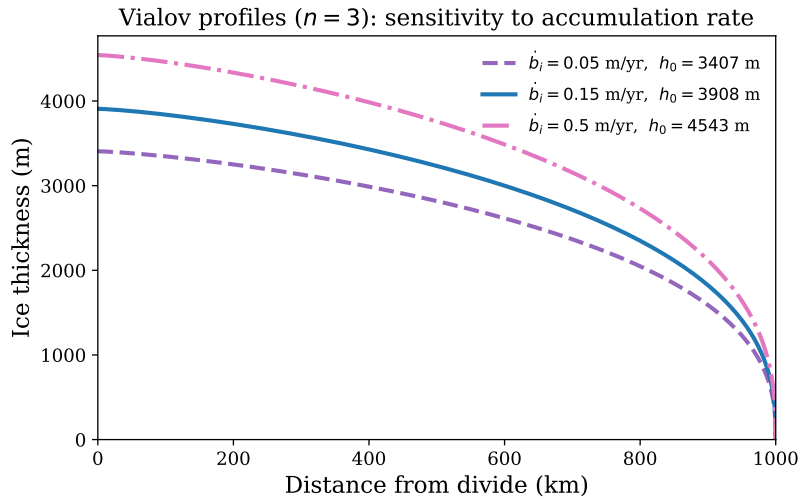

Figure 4: Vialov profiles ($n = 3$) for accumulation rates spanning an order of magnitude. Despite large differences in $\dot{b}_i$, the profiles are strikingly similar.

6 The Perfectly Plastic Limit

An instructive limiting case is obtained by taking $n \to \infty$ in Glen's law. In this limit, ice behaves as a **perfectly plastic material**: it does not deform at all below a yield stress τ_0, and it flows freely at stress equal to τ_0.

If the basal shear stress equals the yield stress everywhere, the force balance gives:

$$\rho_i g\, h \left| \frac{dh}{dx} \right| = \tau_0. \tag{36}$$

Separating variables:

$$h\, dh = -\frac{\tau_0}{\rho_i g}\, dx, \tag{37}$$

where the sign accounts for $dh/dx < 0$ (surface slopes downward from divide to margin). Integrating from the margin ($x = L$, $h = 0$) inward:

$$\boxed{h^2(x) = \frac{2\tau_0}{\rho_i g}\,(L - x).} \tag{38}$$

This is a **parabolic profile**, first derived by [4]. The divide thickness is:

$$h_0 = \sqrt{\frac{2\tau_0 L}{\rho_i g}}. \tag{39}$$

With $\tau_0 \approx 100$ kPa, $L = 1000$ km, and $\rho_i g \approx 9000\ \mathrm{N\,m^{-3}}$:

$$h_0 = \sqrt{\frac{2 \times 10^5 \times 10^6}{9 \times 10^3}} \approx 4700\ \mathrm{m}, \tag{40}$$

which is in the right ballpark for a large ice sheet (the East Antarctic ice sheet has a divide thickness of ∼4000 m for a comparable half-span), though the agreement is best when a more realistic yield stress is chosen.

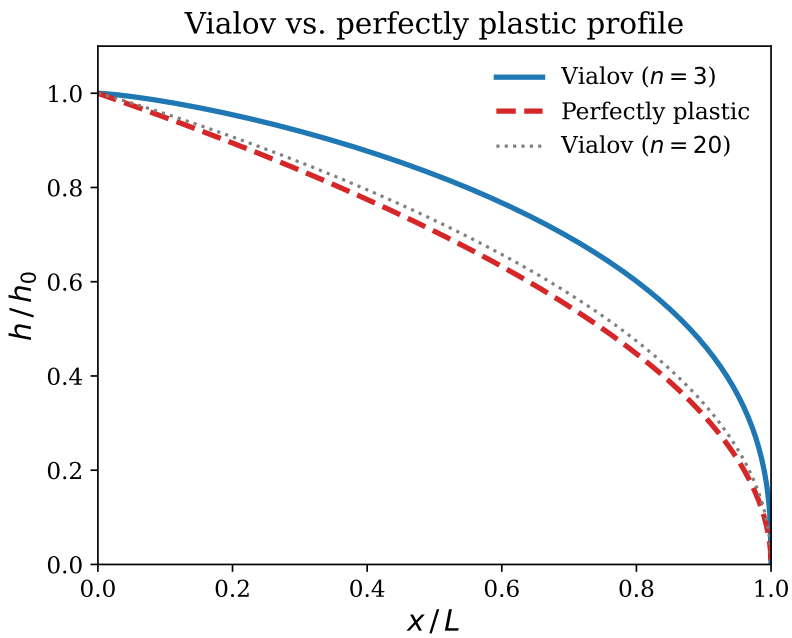

Figure 5: Comparison of the Vialov profile ($n = 3$) with the perfectly plastic (parabolic) profile. The two are remarkably similar: flat in the interior, steep at the margins. The plastic profile slightly overestimates thickness near the divide and underestimates it near the margins.

The resemblance between the Vialov and parabolic profiles (Fig. 5) is striking. Both share the same qualitative features:

- Flat interior (small surface slopes near the divide).
- Steep margins (large slopes as $h \to 0$).
- Divide thickness that scales as $\sqrt{L}$.

This similarity reinforces the conclusion that ice sheet shape is controlled primarily by the geometry and gravity, not by the details of the rheology.

7 Comparison with Observations

How well do these idealized profiles match real ice sheets? Figure 6 compares the Vialov and plastic profiles with a representative transect across the East Antarctic ice sheet.

Interior ($|x| \lesssim 0.7L$): The agreement is generally good. In the slow-flowing interior, the bed is often frozen, basal sliding is negligible, and the SIA assumptions are well satisfied.

Margins ($|x| \gtrsim 0.7L$): The predicted profiles systematically deviate from observations. The ice is typically thinner and the surface steeper than predicted. Several mechanisms contribute:

1. **Basal sliding.** The models assume $u_b = 0$. Near the margins, the bed is often at the pressure melting point, and sliding can be the dominant component of velocity. Sliding evacuates ice more efficiently, producing a thinner profile.

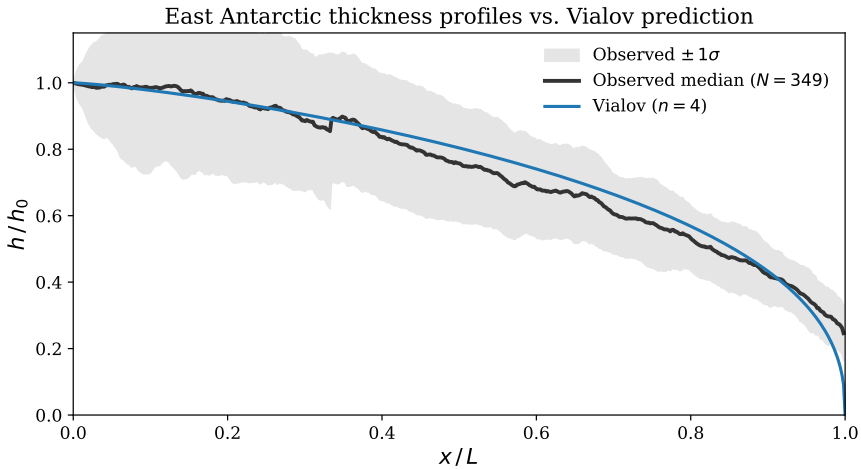

Figure 6: Normalized ice thickness profiles along 349 flowlines in East Antarctica (BedMachine Antarctica v4.1; [2]) compared with the Vialov profile for $n = 4$. Flowlines were seeded in fast-flowing ice and traced upstream to the divide; only profiles reaching the ridge are shown. For each profile, the Vialov half-span L was fitted to minimize the RMS misfit at $n = 4$ (which fits marginally better than $n = 3$), and thickness is normalized by the divide value h_0. The dark line shows the observed median; the shaded band is ± 1 standard deviation. The Vialov prediction captures the interior profile shape well but overpredicts thickness near the margins, where basal sliding and ice stream dynamics — not captured by the SIA — become important.

2. **Ice streams and outlet glaciers.** Fast-flowing features violate the SIA because longitudinal stress gradients and lateral drag are dynamically important. The flow is no longer controlled by local thickness and slope alone.

3. **Temperature variations.** The rate factor A increases with temperature. Since ice near the margins is generally warmer (due to both surface climate and strain heating), the ice there is effectively softer than assumed by a uniform A.

4. **Non-uniform accumulation.** Accumulation rates are not spatially uniform; they tend to decrease with distance from the coast (orographic effects) and vary with elevation.

5. **Marine-terminating margins.** Many Antarctic outlet glaciers terminate in the ocean, where the dynamics of calving, buttressing by ice shelves, and ocean-induced basal melting govern the margin position—none of which is captured by the simple models.

The systematic pattern of failure at the margins is informative: it tells us precisely which physical processes we need to incorporate to build more complete models. The full conservation of linear momentum—and the richer dynamics it permits—is the subject of subsequent lectures.

8 Summary

Key results from this lecture:

1. **The strain-rate tensor** $\dot{\varepsilon}_{ij}$ measures the rate of deformation. It is the symmetric part of the velocity gradient tensor. Incompressibility requires $\dot{\varepsilon}_{kk} = 0$.

2. **Glen's flow law** $\dot{\varepsilon}_{ij} = A\,\tau_e^{n-1}\,\tau_{ij}$ with $n \approx 3$ is the constitutive relation for glacier ice. It implies shear-thinning: deformation concentrates where stress is large.

3. **The parallel-slab solution** gives the velocity profile under simple shear:

$$u(z) = u_b + \frac{2A}{n+1}(\rho_i g \sin\alpha)^n \left[h^{n+1} - (s-z)^{n+1}\right].$$

   For $n = 3$, most shearing occurs in the basal ∼20% of the ice column.

4. **The shallow-ice approximation** treats each column as a local slab, coupling the flux $q \propto h^{n+2}|ds/dx|^{n-1}(ds/dx)$ to the continuity equation.

5. **The Vialov profile** gives the steady-state surface elevation of an ice sheet. Divide thickness depends on accumulation and the rate factor only through $(\dot{b}_i/A)^{1/8}$—an extraordinary insensitivity that reflects the strong nonlinearity of the flux–thickness relation.

6. **The perfectly plastic profile** ($n \to \infty$) is a parabola, $h^2 \propto (L-x)$, and is strikingly similar to the Vialov profile. Ice sheet shape is dominated by geometry and gravity.

7. **Where the SIA fails**—at the margins, in ice streams, and at marine-terminating outlets—tells us what physics must come next: basal sliding, longitudinal stress gradients, lateral drag, and ocean–ice interaction.

Reading

Cuffey & Paterson, *Physics of Glaciers*, 4th ed.:

- Section 3.4: Overview of the flow law (preview for later lectures)
- Sections 8.2–8.3: Strain rates and the velocity gradient tensor
- Sections 8.5.1–8.5.3: Glen's flow law and effective viscosity
- Section 9.3: Parallel-flow solutions (slab on a slope)
- Section 9.4: Steady-state profiles of ice sheets (Vialov)
- Section 9.5.1: Perfectly plastic glaciers

References

[1] Glen, J. W. (1955). The creep of polycrystalline ice. *Proceedings of the Royal Society of London. Series A*, 228(1175), 519–538. `doi:10.1098/rspa.1955.0066`

[2] Morlighem, M., et al. (2020). Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet. *Nature Geoscience*, 13, 132–137. `doi:10.1038/s41561-019-0510-8`

[3] Hutter, K. (1983). *Theoretical Glaciology: Material Science of Ice and the Mechanics of Glaciers and Ice Sheets*. D. Reidel Publishing Company. `doi:10.1007/978-94-015-1167-4`

[4] Nye, J. F. (1952). The mechanics of glacier flow. *Journal of Glaciology*, 2(12), 82–93. doi:10.3189/S0022143000033967

[5] Nye, J. F. (1957). The distribution of stress and velocity in glaciers and ice-sheets. *Proceedings of the Royal Society of London. Series A*, 239(1216), 113–133. doi:10.1098/rspa.1957.0026

[6] Vialov, S. S. (1958). Regularities of glacial shields movement and the theory of plastic viscous flow. *International Association of Hydrological Sciences Publication*, 47, 266–275.